THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Dec 22, 2017 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. With a strong commitment to developing Hong Kong into a smart city, it is expected that a series of smart city related construction projects will be initiated to showcase various emerging smart technologies.

   a) Outline the principle of serial tendering and discuss its suitability for a series of smart city related construction projects. [10 marks]

   b) Describe the mechanism, advantages and disadvantages of negotiation and open tendering and discuss whether the government should adopt either or both of these approaches to solicit tenders for such projects. [15 marks]

2. The schedule and logic of a small-scale construction project are shown in the following table:

Activity	*Duration*	*Predecessor(s)*
A	2	
B	6	A
C	10	A
D	4	A
E	7	B
F	5	B, C
G	3	C, D
H	5	E, F
I	2	G, H

   a) Compute the total float and free float for each activity using the above information. [8 marks]

   b) Based on the results of (a), explain why there could be a difference between the total float and free float and justify the need for calculating the free float from the planning perspective. [5 marks]

   c) Describe with reasons the information and records to be maintained by contractor's site personnel while the work is in progress to support various management decisions. [12 marks]

3. A Hong Kong construction company has just won a bid to build a mega infrastructure project in the "Greater Bay Area" of China.

   a) Highlight the features of functional and project-based organisational setting and explain their pros and cons. [14 marks]

   b) Discuss whether a matrix organisation is more appropriate approach when applied to this scenario. [7 marks]

   c) Prepare an organisation chart based for this project based on the organisational setting you have proposed. [4 marks]

4. a) A consortium is preparing a "build, own, operate and transfer" proposal for a toll road scheme. The project is estimated to cost $500 million and will take two years to construct. In return, the consortium would operate the finished facility for 15 years. It is anticipated that 15,000 vehicles will be commuting through the proposed road every day if the toll is set at $25 per vehicle per journey. At this traffic volume, the road shall be resurfaced every six years at the cost of $100 million. Besdies, a sum of $20 million / annum and $15 million / annum shall be allowed to cover the interest charge as well as the operation and routine maintenance costs respectively. The toll road will be returned to the government at no charge after the service agreement comes to an end. Calculate the internal rate of return for this scheme. [18 marks]

b) While multiple consortia are interested in submitting proposals for this toll road scheme, discuss what factors should the government take into account when awarding a project under the "build, own, operate and transfer" arrangement? [5 marks]

c) On the basis of the government's consideration in (b), justify whether it would be a good idea for the consortium to put forward a higher toll to cover the risk of fall in traffic volume. [2 marks]

5. a) With reference to the roles and responsibilities of civil engineers, explain how civil engineers can serve competently, collaboratively and ethically as master planners, designers, constructors and operators of society's economic and social engine. [17 marks]

b) Discuss with examples what unethical behaviours consultant engineers should be aware of when tenderers are selected from the approved list of contractors. What action(s) should be taken when these unethical behaviours are discovered? [8 marks]

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